IEEE 754 Single-Precision (32-Bit) Format

32-Bit Field Partitioning

- 1-bit Sign (S): 0 for positive values, 1 for negative values.
- 8-bit Biased Exponent (E'): stored as E + Bias with Bias = 127 (range 1..254 normalized).
- 23-bit Fraction / Mantissa (M): fractional bits after the implicit hidden leading 1.

General Value Equation

- Normalized floating-point value = (-1)^S * (1.Fraction)_2 * 2^(E' - 127).
- All-zeros exponent encodes 0 or subnormals; all-ones exponent encodes Infinity or NaN.

Worked Encoding Example: -12.625 to 0xC14A0000

Binary & Normalization Steps

- Sign S = 1 (negative); 12 = 1100_2 and 0.625 = 0.101_2 -> 1100.101_2.
- Shift binary point left 3 places: 1.100101 * 2^3 (Unbiased exponent E = 3).

Exponent Packing & Hex Result

- Biased exponent = 127 + 3 = 130 = 10000010 in 8-bit binary.
- 23-bit fraction = 10010100000000000000000 -> Full hex word = 0xC14A0000.